Virtual Threads

Fan-out 10k blocking HTTP checks

Season 5 · Concurrency · Intermediate

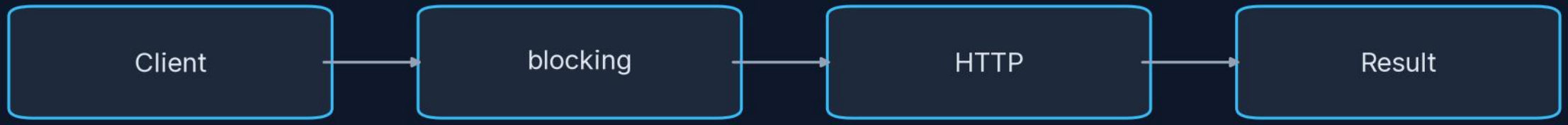
Today's Hook

- Problem: Fan-out 10k blocking HTTP checks
- Naive pain cue: Failure path is asserted in ProblemTest (injected fault / invalid input)
- We will refactor to Virtual Threads with a runnable Demo.
- Failure path is locked by ProblemTest.

The Problem

- Scenario: Fan-out 10k blocking HTTP checks
- Constraints: solve «Fan-out 10k blocking HTTP checks» with Virtual Threads — keep the demo runnable and the failure path observable.
- Naive pain: Naive version fails to cleanly solve: Fan-out 10k blocking HTTP checks
- Success looks like: 200

Structure



Virtual Threads: Fan-out 10k blocking HTTP checks

Real-World Scenario

- Running example: Fan-out 10k blocking HTTP checks
- Key types: UrlChecker
- Happy-path stdout cue: 200
- Failure-path stdout cue: Failure path is asserted in ProblemTest (injected fault / invalid input)

Naive Code

```
package com.buildwithwaqar.patterns.concurrency.virtualthreads.problem;

import java.util.*;

public final class NaiveChecks {
  public static int run(int n){ int ok=0; for(int i=0;i<n;i++) ok++; return ok; }
}
```

Why Naive Hurts

- Naive version fails to cleanly solve: Fan-out 10k blocking HTTP checks
- Without Virtual Threads, call sites absorb the wrong details.
- Failure mode to show on camera: Failure path is asserted in ProblemTest (injected fault / invalid input)
- New requirements force shotgun edits across the naive code.

Solution

```java
package com.buildwithwaqar.patterns.concurrency.virtualthreads.problem;

import java.util.*;
import java.util.concurrent.*;
import java.util.concurrent.atomic.*;
import java.util.concurrent.locks.*;
import java.util.function.*;

final class UrlChecker {
  int checkAll(int n) throws Exception {
    try (var scope = Executors.newVirtualThreadPerTaskExecutor()) {
      List<Future<Boolean>> fs=new ArrayList<>();
      for (int i=0;i<n;i++) fs.add(scope.submit(() -> true));
      int ok=0; for (var f: fs) if (f.get()) ok++;
      return ok;
    }
  }
}
```

Runnable main

```
package com.buildwithwaqar.patterns.concurrency.virtualthreads.problem;
import java.util.*; import java.util.concurrent.*; import java.util.concurrent.atomic.*;
public final class ProblemDemo {
  public static void main(String[] args) throws Exception {
    System.out.println("=== Virtual Threads: fan-out HTTP checks ===");
    System.out.println("--- naive ---"); System.out.println(NaiveChecks.run(200));
System.out.println("--- virtual threads ---");
System.out.println("checks="+new UrlChecker().checkAll(200));
  }
}
```

Walkthrough Output

```
=== Virtual Threads: fan-out HTTP checks ===
--- naive ---
200
--- virtual threads ---
checks=200
```

Step-by-Step Trace

- Match stdout to stages — look for: 200
- Then the injected failure: Failure path is asserted in ProblemTest (injected fault / invalid input)
- Re-run: java -cp … 40-virtual-threads ProblemDemo (see episode folder).
- Pause the video and predict the next printed line.

Modern Java 25 / Zazr

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```

Trade-offs

- Use Virtual Threads when «Fan-out 10k blocking HTTP checks» repeats and a simpler feature does not erase it.
- Refuse extra types when a sealed switch or a function value already suffices.
- Teaching demos omit production metrics, retries, and multi-region clocks.
- Interview soundbite: state the force behind Virtual Threads in product language.

Related Patterns

- Name the pattern usually confused with Virtual Threads and separate the forces.
- Sketch the roles for «Fan-out 10k blocking HTTP checks» from memory in sixty seconds.
- Point to the next episode in this season on the playlist map.
- Link this episode's series/ folder in the YouTube description.

Recap

- Problem: Fan-out 10k blocking HTTP checks
- Naive pain → Virtual Threads boundary → Demo stdout.
- Remember the failure cue: Failure path is asserted in ProblemTest (injected fault / invalid input)
- Break a ProblemTest assertion, watch it fail, restore it.